MILITARY SPECIFICATION

MIL-R-55256A(EL)
5 April 1965
Supersedes
MIL-R-55256(EL)
18 February 1964

RADIO SET AN/ARC-54( )
(UNITS OF)

1 SCOPE

1.1 This specification covers the following units of Radio Set AN/ARC-54( ), one type of multi-channel frequency modulated, airborne radio operating throughout the range of 30.00 to 69.95 megacycles, (See 6.1 and 6.3):

Receiver-Transmitter, Radio RT-348( )/ARC-54
Control, Radio Set C-3835( )/ARC-54
Mounting MT-1535( )/ARC-54

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids (or request for proposal) form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-1	Electron Tubes and Crystal Rectifiers
MIL-P-116	Preservation, Methods of
MIL-T-713	Twine and Tape, Lacing and Tying (For Use in Electrical and Electronic Equipment)
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-I-6181D	Interference Control Requirements, Aircraft Equipment
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-19500	Semiconductor Devices
MIL-M-13231	Marking of Electronic Items

FSC-5821

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STANDARDS

FEDERAL

No. 595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

DRAWINGS

SIGNAL CORPS

SC-GL-323153	Electrical Test Fixtures for Testing AN/ARC-54( )
SC-GL-323193	Master Gage List for Mechanical Gages, AN/ARC-54( )
SC-A-362100A	Preparation of Preproduction Packs
SC-DL-506297	Mounting MT-1535( )/ARC-54
SC-DL-506298	Control, Radio Set C-3835( )/ARC-54
SC-DL-506299	Receiver-Transmitter, Radio RT-348( )/ARC-54

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Radio Set AN/ARC-54( ) is a light weight multi-channel airborne radio set capable of transmitting and receiving frequency modulated voice communications on any one of 800 discrete channels spaced 50 kc throughout the frequency range of 30.00 to 69.95 mc. Frequency selection for the transmitter and receiver is remotely controlled and the selected frequency displayed on the operator's control box. Retransmission and homing facilities are provided. The equipment operates from an aircraft power supply having a nominal voltage of 27.5 volts dc.

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3.2 Construction.- The units on order shall be constructed in accordance with the following drawing lists:

SC-DL-506297	Mounting, MT-1535( )/ARC-54
SC-DL-506298	Control, Radio Set C-3835( )/ARC-54
SC-DL-506299	Receiver-Transmitter, Radio RT-348( )/ARC-54

3.3 Preproduction samples.- The contractor shall furnish two preproduction samples for approval of equipment on order if required by the invitation for bids and contract. (See 4.3)

3.4 Cleaning.-

3.4.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Controls.- All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, protective.- The units shall be given protective finish in accordance with Specification MIL-F-14072, and the applicable drawings. The final paint film shall be Color No. 37038 (Lusterless Aviation Black) according to Federal Standard No. 595. (See 4.4)

3.7 Marking.-

3.7.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4) Front panel marking shall be Group 1 as described in that specification.

3.7.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial numbers.- The following units shall be serial numbered:

Mounting MT-1535( )/ARC-54
Control, Radio Set C-3835( )/ARC-54
Receiver-Transmitter, Radio RT-348( )/ARC-54

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3.8 Securing of Parts.-

3.8.1 General.- Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.8.2 Securing of parts by threaded fasteners.- Nylok or other screws with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250° (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.9 Soldering.-

3.9.1 Solder.- Solder shall conform to the equipment drawings.

3.9.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for thinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and Cabling.- Wiring and cabling shall be neat and sturdy.

3.10.1 Cabling.- Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

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3.10.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 Support.- Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (I) during and after subjection of the equipment to specified service conditions or (II) after service or repair of the equipment in a normal manner.

3.10.5 Clearance.- Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 Connections.-

3.10.7.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.11 Tubes, electron, and semiconductors (transistor and diodes).- These electron devices (tubes and semiconductors) shall conform to Specification MIL-E-1 and to MIL-S-19500. (See 4.4)

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3.12 Service conditions (See 4.8). - The equipment shall meet the following service conditions as Class 1A equipment as specified in Specification MIL-E-5400, except as otherwise specified herein.

3.12.1 Temperature (See 4.8.1). -

(a) Operating. - Operable conforming to limits specified from -40 to +132F. Operable from +132F to +150F and no measurement shall deteriorate by more than a factor of 2. The equipment shall be considered Class 1A except when otherwise specified.

(b) Non-operating. - Exposure of 24 hours at -65F, and 4 hours at +150.

3.12.2 Altitude (See 4.8.1.2). -

Operating. - Altitude conditions up to 35,000 feet. At altitude conditions equivalent to 15,000 feet or higher, the maximum temperature shall be +86F.

3.12.3 Relative humidity (See 4.8.2). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This test shall be performed over a 10 day cycle.

3.12.4 Orientation See 4.8.3). - Indefinite storage in any orientation.

3.12.5 Sand and dust (See 4.8.4). - The equipment shall withstand exposure to sand and dust particles as encountered in desert areas.

3.12.6 Vibration (See 4.8.5). - The equipment shall meet the vibration requirements of Specification MIL-E-5400 for combination aircraft and helicopter equipment.

3.12.7 Shock (See 4.8.6). -

3.12.7.1 Benchhandling (See 4.8.6.1). - The equipment shall meet full specification performance after the shock test.

3.12.7.2 Equipment (See 4.8.6.2). - The equipment shall meet the equipment shock test. The equipment shall meet full specification performance after the shock test.

3.12.7.3 Mounting, crash safety (See 4.8.6.3). - The mounting shall meet the crash safety test of MIL-E-5400. Bending and distortion shall be permitted, but there shall be no failure of the attaching joints, and the equipment or dummy load shall remain in place.

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3.12.8 Explosion proof (See 4.8.7). - The equipment shall be explosion proof per Specification MIL-E-5400.

3.13 Electrical requirements, receiver (See 4.9). -

3.13.1 Squelch (See 4.9.1). - When the receiver is just squelched, an RF signal input of not more than 1.2 microvolts shall unsquelch the receiver.

3.13.2 Minimum squelch level (See 4.9.2). - When the squelch adjust control is in fully clockwise position, the signal required to unsquelch the receiver shall be no more than 3 microvolts.

3.13.3 Output when squelched (See 4.9.3). - The receiver shall be considered squelched when the audio output is 45 db below 50 mw across 150 ohms.

3.13.4 Squelch overlap (See 4.9.4). - The input signal level at which the receiver unsquelches and that signal level at which it again remains squelched shall differ by no more than 6 db. This measurement shall be made with the receiver squelch set for 1.2 microvolts.

3.13.5 Tone operated squelch (See 4.9.5). - The tone operated squelch circuit shall permit the receiver (operating in the tone squelch mode) to be unsquelched only when the proper tone is present on the carrier.

3.13.6 Tone squelch sensitivity (See 4.9.6). - An RF input of 1.2 microvolts, or less, modulated as described in 3.13.5 above, shall unsquelch the receiver.

3.13.7 Selectivity (See 4.9.7). -

(a) 30 kc minimum at the 6 db points.

(b) 55 kc maximum bandwidth at the 60 db points.

(c) 100 kc maximum bandwidth at the 80 db points.

(The pass band response ripple shall be no greater than 1.5 db, i.e., the peak to trough ratio shall be no greater than 3 db.)

3.13.8 Sensitivity (See 4.9.8). - An RF signal input of 1.2 microvolts, or less, modulated 1000 cps at ± 10 kc deviation, shall produce a signal plus noise-to-noise ratio of not less than 10 db when measured at the output of the receiver (connected with de-emphasis) when terminated into a 150 ohm load.

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3.13.9 Quieting (See 4.9.9).- An RF signal of 120 microvolts modulated 1000 cps at ± 10 kc deviation shall produce a signal plus noise-to-noise ratio of not less than 40 db when measured at the output of the receiver when delivering 50 milliwatts audio into a 150 ohm resistive load.

3.13.10 Blocking (See 4.9.10).- When a signal having an amplitude of one volt rms and a frequency equal to that to which the receiver is tuned is applied to the receiver input terminals, it shall be possible to obtain a 10 db signal plus noise-to-noise ratio with a 2.0 microvolt input within 250 milliseconds following the removal of the one volt signal. No blocking shall occur in the receiver under push-to-talk operation.

3.13.11 Spurious responses (See 4.9.11).- The response to any signal, including the image frequency, differing in frequency from the center frequency by more than 50 kc, shall be a minimum of 80 db below the level of the center frequency. A test shall be made on all channels when spurious responses less than 80 db are predictable, but as a minimum no fewer than 20 channels shall be tested to include the use of all crystals.

3.13.12 Receiver audio response (See 4.9.12).- The receiver audio power output shall be balanced and terminated in a 150 ohm resistive load. The audio output level shall be measured across the load resistor and set to 50 milliwatts. The audio frequency response of the receiver with de-emphasis disabled shall not vary more than +1 to -3 dbs as the frequency is varied from 300 to 6000 cps.

3.13.13 Limiting (See 4.9.13).- The limiting characteristics of the receiver shall be such that the audio output shall not vary by more than 2 db when the RF input signal is varied from 6.0 microvolts to 10,000 microvolts. The input signal shall be modulated with a 1000 cps tone and deviated ± 10 kc. The audio gain shall remain fixed.

3.13.14 De-emphasis (See 4.9.14).- The de-emphasis network, with internal disabling facility, shall have a time constant of 75 ± 10 microseconds. The audio response of the receiver shall follow that of the de-emphasis network by ± 2 db.

3.13.15 Audio distortion (See 4.9.15).- When the receiver is delivering a level of 50 milliwatts to a 150 ohm resistive load, the total harmonic distortion shall be less than 7 percent. This requirement shall be satisfied from 300 to 6000 cps, and carrier deviation of ± 10 kc. With input levels from 3 to 100,000 microvolts the distance shall not exceed 7%.

3.13.16 De-sensitization (See 4.9.16).- When a composite RF signal consisting of a 9 microvolt carrier modulated ± 10 kc deviation at 1000 cps, together with a pulsed RF signal of 0.1 volt peak, 2 to 10 kc microseconds" duration and 1000 ± 100 pulses per second, is impressed on the input terminals of the receiver, the signal plus noise-to noise ratio shall be NLT 6 db at 50 milliwatts audio output into a 150 ohm load.

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3.13.17 Maximum RF input to receiver (See 4.9.17).- Regardless of the settings of the receiver controls, no damage shall result to any receiver unit from the operation of its companion transmitter or from the presence of an RF signal of 2 volts rms into the receiver antenna terminals.

3.13.18 Noise rejection (See 4.9.18).- The susceptibility of the receiver to noise of an impulse type shall be such that when a composite RF signal, consisting of a 2 microvolt unmodulated carrier together with a pulsed RF signal of 0.1 volt peak, 2 to 10 microseconds" duration and 1000 pulses per second ± 100 pulses a second, is impressed on the input terminals of the receiver, its audio output for any setting of the pulsed oscillator frequency with full receiver volume shall be not more than 20 milliwatts above the level obtained from the 2 microvolt signal only. The pulsed radio frequency carrier shall be adjusted to the resonant frequency of the receiver.

3.13.19 Homing sensitivity (See 4.9.19).- The overall sensitivity of the homing system shall be such that the field strength necessary to provide 2.4 microvolts input to the receiver from the communications antenna shall provide sufficient input to the homing antenna, which shall be indicated by a flag down condition. The above measurements shall be made when equal energies are being received by each homing antenna, indicating an "on course" direction of the aircraft.

3.13.20 Visual homing indication (See 4.9.20).- The homing system shall operate so that movements of the Indicator ID-48/ARN, or equal, needle and flags shall indicate the following conditions:.

(a) Equal signal levels through two 6.0 db pads of not more than 4.8 microvolts shall result in a flag down condition and the vertical needle being in center position.

(b) With 10 microvolts input and the left pad adjusted to 3.0 db (right pad 6.0 db) the vertical needle shall deflect to the left. Right deflection shall exist when the left pad is adjusted for 6.0 db and the right pad is adjusted to 3.0 db. The right deflection shall be within 20 percent of the left deflection.

(c) With signal levels from 10 to 10,000 microvolts and 3.0 db differential between the pads, the needle deflection left or right shall not deflect less than the wing span on the horizontal scale.

3.14 Electrical requirement for transmitter.-

3.14.1 Distortion, audio (See 4.10.1).- With an input sufficient to deviate the carrier ± 10 kc, the audio harmonic distortion present on the carrier modulation shall not exceed 7 percent. This requirement shall be met with a deviation of ± 10 kc modulated from 300 cps to 6000 cps.

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3.14.2 Power output (See 4.10.2).- With the transmitter operating into a 52 ohm resistive load, a minimum of 10 watts of RF power shall be delivered to the load for any variation of service conditions with a transmitter input voltage of 27.5 volts dc measured at the input terminals of the transmitter.

3.14.3 Carrier noise (See 4.10.3).- The audio noise level on the carrier, with the audio input terminated by a 150 ohm resistor, shall be a minimum of 40 db below the level of a standard test signal. Only that noise contained in the frequency spectrum below 6000 cps shall be measured. The measurement shall be made through a discriminator and a de-emphasis network having the same characteristics as those employed in the receiver. The 150 cycle tone shall be disabled.

3.14.4 Sidetone (See 4.10.4).- When the carrier is fully modulated the sidetone output shall be not less than 8 db below the receiver output. Total harmonic distortion of the sidetone output shall be 7 percent or less. Sidetone fidelity shall be within +3 to -5 db of the level at 1000 cps for any frequency from 300 to 6000 cps.

3.14.5 Frequency deviation (See 4.10.5).- An input of 0 dbm at 1000 cps shall cause the transmitter to be frequency modulated with ± 10 kc deviation. Deviation shall be adjustable within the following limits by means of internal adjustment: maximum deviation of NLT 13.5 kc; minimum deviation setting shall result in a deviation of NLT 6.5 kcs. This characteristic shall be maintained throughout the 800 channels.

3.14.6 Tone deviation (See 4.10.6).- Operation of the tone of 150 cps ± 3 cps shall deviate the transmitter 3 kc ± 500 cps. This tone shall be transmitted when the squelch control is in DIS, CARRIER or TONE positions and the function switch in "PTT-RETRAN or HOME" position.

3.14.7 Frequency response (See 4.10.7).- With the preemphasis network disabled and the audio input level to the transmitter constant at 0 dbm, the deviation produced by a modulating frequency of 300 to 6000 cps shall be within ± 2.0 db of the deviation produced by a standard test signal.

3.14.8 Preemphasis (See 4.10.8).- The preemphasis RC network with disabling facility shall have a time-constant of 75 microseconds ± 10 percent. The audio response of the transmitter with preemphasis shall follow that of the preemphasis network within ± 2 db.

3.14.9 Retransmission (See 4.10.9).- When two AN/ARC-54 systems are connected for retransmit condition an RF input of 1.5 microvolts deviated ± 10 kc at 1000 cps shall cause the non-receive system to have a power output containing 1000 cps modulation.

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3.15 Frequency Selection (See 4.11).- The time required to effect a change to the frequencies as specified shall not exceed five seconds. An 800 cps tone shall be present at the receiver output during channeling.

3.16 Voltage variations (See 4.12).- The injection frequency of the equipment shall not deviate more than ± 0.0035 percent from the frequencies specified when the input voltage is changed from 24 to 29 volts dc.

3.17 Frequency stability (See 4.13).- The center frequency of the receiver section and the output frequency of the transmitter shall be within ± 5 kc of the nominal frequency within five minutes of warm-up time. This requirement shall be met with any combination of service conditions.

3.18 Preconditioning (See 4.5 and 4.6).-

3.18.1 Electrical preconditioning (See 4.6.1).- Each unit on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the electrical preconditioning.

3.18.2 Bounce preconditioning (See 4.6.2).- Each unit on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to bounce preconditioning for two minutes.

3.19 Radio interference suppression (See 4.14).- The equipment shall meet the requirements of MIL-I-6181, except for the test limits and methods specified in this specification.

3.20 Control, Radio C-3835( )/ARC-54 (See 4.15).- The control shall select the operating frequencies of the RT unit as well as control the ON-OFF, squelch, volume control, etc.

3.21 Operational inspection (See 4.16).- The equipment shall be capable of transmitting and receiving on all channels from 30 to 69.95 mc. The performance meter shall indicate readings between 20 and 70 microamperes at positions 1 through 10, less than 8 microamperes at position 11, and zero at position 12 during applicable mode of operation.

3.22 Operational test (500 hour)(See 4.17).- Receiver-Transmitter RT-348/ARC-54 shall be capable of 500 hours of continuous operation without failure or degradation of performance below established specification limits. During the 500 hour test the following conditions shall apply:

(a) No signal input to receiver.

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(b) The frequency of the RT unit shall be changed 80 times per day every megacycle from 30 to 69 and return to 30 each 24 hour day.

(c) The actual hours of operation shall be determined by use of an elapsed time meter at the test facility.

(d) A daily check on power output and signal plus noise-to-noise ratio at 30, 50 and 69 megacycles.

(e) At the conclusion of the 500 hour period perform the tests specified.

3.23 Interchangeability (See 4.19).- Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. Individual items shall not be handpicked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 Mounting MT-1535( )/ARC-54 (See 4.8.6.3).- The mounting shall meet the service conditions as specified herein.

3.25 Technical literature and spare parts.- Technical literature and spare parts shall be furnished as specified in the contract. Spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.26 Workmanship.- The units shall be manufactured and assembled in accordance with the applicable drawings and the following paragraphs:

Cleaning	3.4
Controls	3.5
Securing of parts	3.8
Soldering	3.9
Wiring and cabling	3.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (See 4.3).

(b) Inspection covered by subsidiary documents (See 4.4).

(c) Quality conformance inspection:

(1) Quality conformance inspection of preparation for delivery (See 4.20).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (See Tables II, III and IV, respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, benchhandling, (3) shock, equipment; other preproduction inspection may precede, follow or be interspersed between the foregoing.

Table I. - Preproduction Inspection

Inspection	Rqt Para	Test Para
Orientation	3.12.4	4.8.3
Sand and dust	3.12.5	4.8.4
Benchhandling	3.12.7.1	4.8.6.1
Equipment	3.12.7.2	4.8.6.2
Mounting (crash safety)	3.12.7.3, 3.24	4.8.6.3
Explosion proof	3.12.8	4.8.7

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where required
Finish, protective	3.6
Marking	3.7
Tubes and semiconductors	3.11

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. All units on order shall be subjected to preconditioning (See 3.18) after final assembly. The preconditioning (3.18) shall be performed once, except when the units require reprocessing.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.16) shall be last (See 6.5).

Table II. - Group A Inspection

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.26	4.18		
Modules:				
Line filter			0.65%	2.5%
Power supply			"	"
Homer			"	"
Mechanical drive			"	"
Harmonic filter			"	"
Variable IF			"	"
RF amplifier			"	"
Power amplifier			"	"
Transmit audio			"	"
Receiver audio			"	"
Sel-call			"	"
RF chassis			"	"
Low frequency oscillator			"	"
High frequency oscillator			"	"
Fixed IF			"	"
RT-348( )/ARC-54 (assembly			1.0 %	4 0%
Mounting MT-1535( )/ARC-54			"	"
Control, Radio C-3835( )/ARC-54			"	"

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Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Electrical				
Transmitter:				
Distortion, audio	3.14.1	4.10.1	1.0 %	*
Power output	3.14.2	4.10.2	for	
Carrier noise	3.14.3	4.10.3	combined	
Sidetone	3.14.4	4.10.4	group	
Frequency deviation	3.14.5	4.10.5		
Tone deviation	3.14.6	4.10.6		
Retransmission	3.14.9	4.10.9		
Receiver:				
Squelch	3.13.1	4.9.1		
Tone operated squelch	3.13.5	4.9.5		
Tone squelch sensitivity	3.13.6	4.9.6		
Sensitivity	3.13.8	4.9.8		
Quieting	3.13.9	4.9.9		
Limiting	3.13.13	4.9.13		
Audio distortion	3.13.15	4.9.15		
Noise rejection	3.13.18	4.9.18		
Homing sensitivity	3.13.19	4.9.19		
Visual homing indication	3.13.20	4.9.20		
Selectivity	3.13.7	4.9.7		
Frequency selection	3.15	4.11		
Control, Radio C-3835( )/ARC-54	3.20	4.15		
Operation inspection	3.21	4.16	0.1 %	

* All electrical defects are considered major.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection.

4.5.3 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

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Table III. - Group B Inspection

Inspection	Rqt Para	Insp Para	AQL
Transmitter:			
Frequency response	3.14.7	4.10.7	6.5 % for combined group
Preemphasis	3.14.8	4.10.8	
Receiver:			
Minimum squelch level	3.13.2	4.9.2	
Output when squelched	3.13.3	4.9.3	
Squelch overlap	3.13.4	4.9.4	
Audio response	3.13.12	4.9.12	
Deemphasis	3.13.14	4.9.14	
Maximum RF input to receiver	3.13.17	4.9.17	
Voltage variation	3.16	4.12	
Frequency stability	3.17	4.13	
Interchangeability (mechanical)	3.23	4.19	6.5 %
Interchangeability (module performance)	3.23	4.19	6.5 %

4.5.4 Group C inspection.- This inspection shall be as listed in Table IV and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. No defects shall be allowed.

Table IV. - Group C Inspection

Inspection	Rqt Para	Insp Para
Subgroup I:		
Blocking	3.13.10	4.9.10
Spurious response	3.13.11	4.9.11
Desensitization	3.13.16	4.9.16
Temperature	3.12.1	4.8.1.1
Altitude	3.12.2	4.8.1.2
Subgroup II:		
Relative humidity	3.12.3	4.8.2
Vibration	3.12.6	4.8.5
Radio interference suppression	3.19	4.14
Operational inspection (500 hours)	3.22	4.17

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4.5.4.1 Sampling for inspection of equipment.- Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.4.1.1 Subgroup I.- For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.4.1.2 Subgroup II.- For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 250, or two units each three months if less than 250 are produced, shall be selected.

4.5.5 Noncompliance.- If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5.6 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Preconditioning.-

4.6.1 Electrical preconditioning (20 hours run-in).- Each RT unit on order shall be electrically preconditioned by subjection to a 20-hour run-in period. Prior to use on contract, the preconditioning test fixture shall be approved by the contracting officer. The unit connected to the approved test fixture shall be operated continuously for 20 hours without failure. The duty cycle shall be transmit for 15 seconds, receive for 45 seconds, and frequency selection shall be performed once every 15 minutes on at least 6 channels. The transmitter shall be terminated into a dummy load.

4.6.2 Bounce preconditioning.- Place the units, with isolators removed or blocked, in their normal operating position on a vibration table. The vibration system shall operate at 55 cps vibration giving vertical excursion sufficient to maintain an acceleration of 55 G's.

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b. Mechanical operation of moving parts such as

4.7 Standard test conditions.- Unless otherwise specified herein, all inspections shall be performed at room temperature, pressure and humidity under the following standard conditions:

(a) Power supply voltage.- 27.5 volts dc ± 0.5 volts.

(b) Operational duty cycle.- One minute transmit 10 minutes receive.

(c) Warm-up time.- Not more than 5 minutes.

(d) Output impedance.- RF 52 ohms, audio 150 ohms.

(e) Output level.- RF 10 watts, audio 50 mw

(f) Input impedance.- RF 52 ohms, audio 150 ohms.

(g) Input level.- Audio 0 dbm at 1000 cps.

4.7.1 Standard test signal.- The standard test signal shall be a frequency modulated carrier deviated ± 10 kcs at a 1000 cps rate produced by a signal generator having an internal impedance of 52 ohms. A 52 ohm 6 db (antenna simulator) pad shall be inserted between the signal generator output terminals for applicable measurements. Signals shall be measured at input to the pad.

4.8 Service condition tests.- The service condition tests shall be conducted in accordance with MIL-T-5422 for Class 1A equipment as defined in Specification MIL-E-5400. Electrical tests shall be run to verify compliance with Paragraph 3.14.2 and 3.13.8 at carrier frequencies of 30.00, 50.00 and 68.00 mc.

4.8.1 Altitude and temperature.- Using a power source of 29 volts the following tests shall be performed.

4.8.1.1 Temperature.-

(a) Operating.- From -40F to +132F.

(b) Non-operating.- From -65F to +150F.

4.8.1.2 Altitude.- Sea level to 35,000 feet above sea level (operating) for conditions equivalent to 15,000 feet or higher, the maximum temperature shall be +86F.

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4.8.2 Relative humidity.- The equipment shall be subjected to the humidity test of Specification MIL-T-5422.

4.8.3 Orientation.- The equipment shall be inclined for a minimum of five minutes in each plane (forward, backward, left and right) to an angle of 20 ± 3 degrees. During inclination in each plane the equipment shall meet specified performance for the operational inspection of 4.16.

4.8.4 Sand and dust.- The equipment shall be subjected to the sand and dust test of Specification MIL-T-5422.

4.8.5 Vibration.- The equipment shall be subjected to the vibration test of MIL-T-5422 for compliance with Curve IV of Table 2. Tests shall be performed before, during, and after vibration to insure compliance with Paragraphs 3.14.2 and 3.13.8 at supply voltage of 27.5 volts and carrier frequencies of 30.00, 50.00 and 68.00 mc.

4.8.6 Shock.-

4.8.6.1 Benchhandling.- The chassis and front panel assembly shall be removed from the enclosure, as for servicing, and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of four drops.

(b) Repeat (a), with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before and after shock to insure compliance with Paragraphs 3.14.2 and 3.13.8 at supply voltage of 27.5 volts and carrier frequencies of 30.00, 50.00 and 68.00 mc.

4.8.6.2 Equipment.- The equipment shall be subjected to the shock tests of Specification MIL-T-5422.

4.8.6.3 Mounting (crash safety).- The equipment shall be subjected to the test of Specification MIL-T-5422.

4.8.7 Explosion proof.- The equipment shall be subjected to the test of Specification MIL-T-5422 except as limited by Paragraph 3.12.2.

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4.9 Receiver, Electrical requirements.-

4.9.1 Squelch.- Under standard test conditions and with standard test signal set for receive operation, adjust the squelch control with no RF input until receiver is just squelched. Measure the point where the receiver is just unsquelched as RF input is increased. Perform test at 50.50 mc. Requirements of 3.13.1 shall be met.

4.9.2 Minimum squelch level.- Adjust squelch control for minimum sensitivity (fully clockwise) and measure RF input required to unsquelch the receiver. Make measurement at 31.00 and 68.95 mc. Requirements of 3.13.2 shall be met.

4.9.3 Output when squelched.- Set RF input to 0.6 microvolts deviated 10 kc at 1000 cps and with squelch control adjusted for minimum sensitivity (fully clockwise) measure receiver audio output relative to 50 milliwatt reference. Make measurement at 31.00 and 68.95 mc. Requirements of 3.13.3 shall be met.

4.9.4 Squelch overlap.- With ARC-54( ) set for receive operation and in "CARR" squelch position, adjust squelch with no RF signal input so that receive audio output is just squelched. Apply RF signal input deviated ±10 kc at 1000 cps. Measure the RF input at the point where the receiver unsquelches as the RF input is increased from 0.1 microvolt. Measure the point where the receiver again squelches as the RF input is reduced. Make measurements at 31.00, 41.75, 50.50 and 68.95 mc and at supply voltages of 24, 27.5 and 29 vdc. Requirements of 3.13.4 shall be met.

4.9.5 Tone operated squelch.- Under standard test conditions with the RT-348( )/ ARC-54 set for TONE squelch receive operation, adjust the RF input to 100 microvolts deviated ±2500 cps at a modulating frequency of 150 cps measure the frequencies which produce squelch and unsquelch as the modulating frequency is varied from 130 cps to 170 cps. Repeat test with deviation of ±3500 cps. Perform tests at 50.50 mc. The requirements of 3.13.5 shall be met.

4.9.6 Tone squelch sensitivity.- Under standard test conditions with ARC-54( ) set for TONE squelch receive operation, apply an RF input of 1.2 microvolts deviated ±3000 cps at a modulating frequency of 150 cps. Verify that this input unsquelches the receiver. Perform test at 50.50 mc for compliance with 3.13.6.

4.9.7 Selectivity.- Under standard test conditions and with the RT-348( )/ARC-54 set for receive operation, measure the bandpass characteristic of the receiver while varying the input frequency. Tests shall be made at 31.00, 50.50 and 68.95 mc for compliance with 3.13.7.

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4.9.8 Sensitivity.- Set generator output level to 1.2 microvolts deviated ± 10 kc at 1000 cps with de-emphasis connected and squelch disabled. With the RT-348( )/ARC-54 set for receive operation, measure and record ratio of audio output when signal is modulated to audio output when signal is unmodulated. Perform tests at 27.5 vdc and at the carrier frequencies listed below. Frequencies marked with an asterisk will be re-checked at 24 vdc and 29 vdc. The requirements of 3.13.8 shall be met.

MC	MC	MC	MC
*31.30	41.30	51.55	61.15
33.10	43.35	53.60	63.80
35.15	*45.40	55.65	65.90
*37.20	47.45	57.70	68.90
39.25	*50.50	*59.35	*69.00

4.9.9 Quieting.- Under standard test conditions with standard test signal in receive operation, set the RF input level to 120 microvolts and measure the signal plus noise-to-noise. Make measurements at the frequencies listed in Paragraph 4.9.8 The requirements of 3.13.9 shall be met.

4.9.10 Blocking.- Under standard test conditions with standard test signal set for receive operation, with squelch in disable position, apply two RF signals, tuned to the center frequency of the receiver, to the input of the receiver. Adjust one RF input for 2 uv ± 10 kc deviation at 1000 cps and the second RF for one volt unmodulated. Measure the time required for the audio output to recover following the removal of the one volt signal. Make tests at carrier frequencies of 32.15, 40.00, 51.65, 59.95 and 64.85 mc for compliance with 3.13.10.

4.9.11 Spurious response.- Under standard test condition with standard test signal check receiver spurious responses from 20 to 150 mc at the frequencies listed below, except the frequencies marked with an asterisk shall be checked from 150 kc to 500 mc. The requirements of 3.13.11 shall be met. During the Group C tests five of the frequencies listed below shall be used. The five frequencies shall be selected during preproduction inspection by the contracting officer.

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MC	MC	MC	MC	MC	MC
*30.25	33.60	36.10	38.80	41.15	45.10
30.30	33.65	*36.45	38.85	41.20	45.65
30.35	34.20	36.50	38.90	41.25	45.70
31.10	34.25	36.55	38.95	41.30	45.80
31.15	34.30	36.60	39.60	42.75	45.85
31.20	34.35	36.70	39.65	42.80	47.25
31.25	*34.40	37.25	39.70	42.85	47.30
31.30	34.90	37.35	39.75	*42.90	47.35
32.00	34.95	37.40	39.80	43.40	47.90
32.05	35.00	37.45	40.10	43.45	51.75
32.10	35.05	38.20	40.50	43.50	51.80
32.60	35.10	38.25	40.55	43.55	51.85
32.65	35.95	38.30	40.60	*44.95	
32.70	36.00	38.35	40.65	45.00	
33.55	36.05	38.40	41.10	45.05	

4.9.12 Audio response.- Under standard test conditions with standard test signals set for receive operation with de-emphasis disabled and RF input of 100 uv deviated ± 10 kc at 1000 cps, adjust the audio level potentiometer for 50 milliwatts. Vary the modulation from 300 to 6000 cps. Measurements shall be made at 31.00 and 68.95 mc with power supply voltages of 24, 27.5 and 29 volts dc. Requirements of 3.13.12 shall be met.

4.9.13 Limiting.- Under standard test conditions, with standard test signal in receive operation, measure the change in audio output from 6 to 10,000 microvolts RF input, at the frequencies marked with an asterisk in 4.9.8. Requirements of 3.13.13 shall be met.

4.9.14 De-emphasis.- Under standard test conditions with standard test signal set for receive operation with de-emphasis and with an RF input of 100 microvolts ± 10 kc deviation, measure the audio response as the modulating frequency is varied through the range of 300 to 6000 cps. Measurements shall be made at carrier frequencies of 31.00, 41.75, 50.50 and 68.95 mc. The requirements of 3.13.14 shall be met.

4.9.15 Audio distortion.- Under standard test conditions with standard test signal set for receive operation with de-emphasis connected, an RF input of 100 microvolts and ± 10 kc deviation, measure the distortion at RF carrier frequencies of 31.00, 50.50 and 68.95 mc, as the modulating frequency is varied from 300 to 6000 cps. The requirements of 3.13.15 shall be met. The RF input levels from 3 to 100,000 microvolts shall be conducted at 50.50 mc with 1000 cps ± 10 kc.

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4.9.16 De-sensitization.- Under standard test conditions with standard test signal set for receive operation, with squelch disabled, apply two RF signals, tuned to the receiver frequency, to the input of the receiver. Apply an input signal of 9 microvolts modulated ± 10 kc deviation at 1000 cps and another input signal 0.1 volt peak pulse modulated with 2 to 10 microseconds' pulse at a rate of 1000 pps ± 100 pps, to the receiver input. With this composite signal applied measure the signal plus noise-to-noise ratio of the receiver at channel frequencies of 31.05, 44.45, 51.25, 59.95 and 68.00 mc, for compliance with 3.13.16.

4.9.17 Maximum RF input to receiver.- Under standard test conditions with standard test signal set for receive operation, apply a 2 volt input to the receiver for one minute. Following this, remove the signal and make measurements to determine that receiver performance has not been degraded. Perform test at 50.50 mc, for compliance with 3.13.17.

4.9.18 Noise rejection.- Using the same measurement technique as above (4.9.16) measure the audio output of receiver produced by pulse type interference. Apply an input signal of 2 microvolts unmodulated and another RF input of 0.1 volt peak pulse modulated with a pulse width of 2 to 10 microseconds and a rate of 1000 pps ± 100 pps to the receiver input. Make measurements at carrier frequencies of 31.15, 39.95, 43.65, 51.80 and 65.50 mc for compliance with 3.13.18.

4.9.19 Homing sensitivity.- Using the same measuring techniques as in 4.9.20, vary the RF input above and below the squelch operating flags. Both vertical and horizontal flags of the ID-48/ARN should appear when the receiver is squelched, and disappear when the receiver is unsquelched. Measure at 50.50 mc and at voltage of 24, 27.5 and 29 volts dc for compliance with 3.13.19.

4.9.20 Visual homing indication.- Under standard test conditions with standard test signal set for receive operation, place the receiver in "HOME" and "CARR" position. From a common generator apply signals to the homing antenna input. Adjust level of the input so that (1) right is stronger than left, (2) left is stronger than right, (3) equal amplitudes are applied to both inputs, and (4) signal strength is varied over the levels specified. Observe the vertical needle of ID-48. Measure at carrier frequencies of 31.00, 50.50 and 68.95 mc for compliance with 3.13.20.

4.10 Electrical requirement (transmitter).-

4.10.1 Distortion, audio.- Using an audio input of 0.37 volts rms at 1000 cps with pre-emphasis, measure the distortion of the transmitted audio at 300, 1000, 2000, 4000, and 6000 cps. Adjust audio input level for 10 kc at each audio frequency. Measurements shall be made at 31.00, 50.50 and 68.95 mc and results shall be in accordance with paragraph 3.14.1. The carrier shall not be deviated by the 150 cps tone during this test.

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4.10.2 Power output.- With standard conditions, the power output of the transmitter shall be checked at the frequencies listed below for compliance with 3.14.2.

MC	MC	MC	MC
30.00	40.25	49.25	60.75
32.05	42.30	52.55	62.50
34.10	44.35	54.60	64.85
36.15	46.40	56.65	66.90
38.20	48.45	58.70	69.95

4.10.3 Carrier noise.- Using standard test conditions with de-emphasis and pre-emphasis, connect the output of an RT-348( )/ARC-54 transmitter through an attenuator to the input of the receiver of another RT-348( )/ARC-54. Set the RF level into the receiver at 250 microvolts. Record the output of the receiver. Remove the modulation and record carrier noise level. Tests shall be conducted at carrier frequencies of 31.00, 50.50, and 68.95 megacycles. The requirements of 3.14.3 shall be met. The carrier shall not be deviated by the 150 cps tone during this test.

4.10.4 Sidetone.- With the RT unit operating as a transmitter measure the sidetone audio output noting its level compared to 50 milliwatts. Modulate the transmitter with standard audio input. When carrier is present, note output of side tone audio, then disable carrier and measure sidetone audio output. Tests shall be conducted at 31.00, 50.50, and 68.95 mc and supply voltages of 24, 27.5 and 29 volts dc, for compliance with 3.14.4.

4.10.5 Frequency deviation.- With standard conditions and with preemphasis disabled, rotate the deviation control from maximum to a minimum position. Make minimum and maximum deviation measurements at 24, 27.5 and 68.95 mc. The transmitter shall meet the requirements of 3.14.5.

4.10.6 Tone deviation.- Under standard conditions and without audio input, place the squelch control in any of the positions stated and measure deviation as the transmitter is keyed. The requirements of 3.14.6 shall be met.

4.10.7 Frequency response.- Set the RT unit to 50.50 mc, disable the preemphasis network, and with standard input level to the transmitter, vary the modulation frequency from 300 cps to 6000 cps and measure the audio response. Results shall be in accordance with 3.14.7.

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4.10.8 Preemphasis.- Using an audio input of 0.37 volts rms at 1000 cps and with 10 kc deviation preemphasis, vary the modulated frequency and input level according to the following chart. Tests shall be performed at carrier frequencies of 31.00, 50.50 and 68.95 mc, for compliance with 3.14.8.

CPS	VOLTS	CPS	VOLTS
300	0.39	2000	0.29
600	0.39	4000	0.18
1000	0.37	6000	0.13

4.10.9 Retransmission.- Connect two AN/ARC-54 systems for retransmit. Set the Function switch of each C-3835( )/ARC-54 to RE-TRANSMIT position and each squelch switch to the same position ("CARR"). Apply a RF signal to the communication antenna terminals of either system. Tests shall be conducted with 40.50 and 60.50 mc, for compliance with 3.14.9.

4.11 Frequency selection.- Using a stopwatch, monitoring the system and the channel changing tone, measure the time required for the RT unit to change channels. Make measurements at the following frequencies and supply voltages of 24, 27.5 and 29 volts, for compliance with 3.14.10:

(a) 30.00 to 69.00 mc
(b) 69.00 to 30.00 mc
(c) 45.00 to 46.00 mc
(d) 46.00 to 45.00 mc
(e) 50.00 to 50.95 mc
(f) 50.95 to 50.00 mc
(g) 61.45 to 61.50 mc
(h) 61.50 to 61.45 mc

4.12 Voltage variation.- Set the megacycle and kilocycle controls to 30.00 mc. Record each 0.05 megacycle step through 0.95 megacycles. Set the control for 30.00 mc. Record each even megacycle through 69.95 mc. The equipment shall met the requirement of 3.16.

4.13 Frequency stability.- Under standard conditions and standard test signal, as applicable, measure the frequency and monitor the transmitter output at 31.00, 50.50 and 68.95 mc for compliance with 3.17.

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4.14 Interference reduction.- The equipment shall be tested in accordance with Specification MIL-I-6181D except as follows:

4.14.1 Conducted interference.- This test to be performed on the power line only. The values indicated in Figure 7, 8, 9 and 10 shall be exceeded by not more than 15 db. (Figures: Reference MIL-I-6181D)

4.14.2 Radio frequency conducted (Par. 4.3.1.1, MIL-I-6181D).- This test shall be conducted on power lines only. The frequency range shall be 0.150 mcs to 30.00 mcs. The signal-to-noise ratio of the receiver shall not be degraded by more than 2 db.

4.14.3 Transmitter key-up or receiver (Par 4.3.3.1, MIL-I-6181D).- The RF output of any transmitter key-up or receiver shall not exceed 80 db above a one-microvolt for CW operation.

4.14.4 Broadband intermodulation (Par. 4.3.4.4.2, MIL-I-6181D).- This test shall not apply.

4.14.5 Two signal intermodulation (Par. 4.3.4.4.1, MIL-I-6181D).- The magnitude of each signal generator shall be 70 db above one microvolt.

4.14.6 Audio frequency conducted (Par. 4.3.4.1.2, MIL-I-6181D).- Substitute the following: This test shall be conducted on power lines only. With one volt rms open circuit applied as in Figure 19 of MIL-I-6181D, the following shall occur: With the squelch control set for 1.2 volts at the operating frequency and with no signal input, the audio output shall be noted. When audio frequencies of 50 to 15,000 cps are applied as in Figure 19, the audio output of the receiver shall be at least 40 db below normal output.

4.14.7 Radiated interference.- The radiated interference shall not exceed the limits established by Paragraph 4.3.2 of MIL-I-6181D by more than 10 db. This test shall be conducted on carrier frequency.

4.14.8 Susceptibility (Par. 4.3.4.2, MIL-I-6181D), radio frequency radiated.- Not more than 3 db of receiver quieting shall be produced when the equipment is subjected to the radio frequency field of this test.

4.15 Control, Radio C-3835( )/ARC-54.- Verify that the FUNCTION and SELECTOR switches and VOLUME control operate the RT unit and meet the requirements of 3.20.

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4.16 Operational inspection.- Each receiver-transmitter shall be combined with corresponding components and associated equipment to form an operational unit. (Substitution of simulated associated equipment is permissible when operational conditions are accurately duplicated. Test fixtures shall be approved by the contracting officer.) Power shall be supplied at low line voltage (24 volts dc). Frequency selection shall be such that all crystals are activated at least once. The requirements of 3.21 shall be met.

4.17 Operational test (500 hour).- At the conclusion of the 500 hour test, the equipment shall be subjected to and meet the requirements listed in Tables II and III.

4.18 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in MIL-STD-252.

4.19 Inspection for interchangeability.- The mechanical dimensions and electrical performance required by the mechanical and electrical test fixture drawings shall be measured to determine conformance with the physical and functional interchangeability requirements of 3.23. When a listed dimension or performance requirement is not within specified limits it shall be considered a major defect. The following dimensions or performance shall be gaged or reasured:

(a) Mechanical (dimensions).- The gage points covered by the following gage list shall apply: SC-GL-323193, Master Gage List for Mechanical Gages for Radio Set AN/ARC-54( ).

(b) Electrical performance (modules).- Prior to assembly into the units, modules shall be tested to determine compliance with the applicable performance requirements specified in the following gage list and associated drawings: SC-GL-323153, Electrical Test Fixtures for Testing Radio Set AN/ARC-54( ).

4.20 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.21 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (See 6.2(e)), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling: 4.16, Operational.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

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5.1.1 Level A.- Radio Set AN/ARC-54( ) shall be preserved, packaged and tested in accordance with the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration, and damage during worldwide shipment handling, and open storage. Methods of packing shall be as follows:

Component	Method of Packaging
Rec/Trans	IIB
Mounting	IC5
Control	IIB
Tech Manuals	IC6

5.2 Level C.- Radio Set AN/ARC-54( ) shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.3 Pilot pack.- When Level "A" packaging and packing is specified above, one acceptable model of Radio Set AN/ARC-54( ) will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100A.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Radio Set AN/ARC-54( ),when installed in aircraft, provides two-way radio communication by means of FM radio telephony (F3), homing on any radio signal of sufficient signal strength within the frequency range of 30.00 to 69.95 mc, retransmission in an automatic, unattended radio repeater type of operation.

6.2 Ordering Data.-

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

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(c) Level of packaging and level of packing required for shipment (Level A, or Level C).

(d) The specific paragraphs of Section 5, which are applicable to the particular procurement.

(e) Preproduction inspection, preproduction pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted.
Inspection to be performed thereon (including rough handling).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and spare parts required.

6.3 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/ARC-54W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending decision from the contracting officer on the adequacy of corrective action (See 4.5.5).

6.5 Location of operational inspection.- It is desirable that the operational inspection (4.16) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army EL

Preparing activity:
Army EL
Project No. 5821-A017

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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